

## 1 / THE PROBLEM

# Missed 15 minutes? Catch up.

RaceTime Copilot helps you understand a chosen race interval through timestamped evidence.



## The problem

Long broadcasts are hard to catch up on. Scrubbing can miss key moments or reveal future results.

## The useful result

Choose a window, follow a runner, and see what the sources reported - including disagreements.

## What works today

Video jobs, learned retrieval and cited recaps. A key-free evidence demo is also included. See validation limits.

## 2 / WHO IT HELPS

# For the people following the race.

Built by Siva Babu: ultramarathoner, race organizer and observability practitioner.



## Fans and families

Catch up together on a missed interval. Inspect the evidence behind a reported sighting or position.

## Organizers and crews

Review imported commentary and timing context. Keep uncertain reports visible.

## Why this project

A personal race-viewing problem meets systems skills: time boundaries, memory limits and useful traces.

3 / HOW IT WORKS

# One question. One evidence workflow.

Save a video, choose an interval and spoiler cutoff, then ask what happened. Review the cited evidence.



**Retrieve**

Keep only evidence fully inside the window.

Inspect bounded video windows and rank learned evidence vectors.

**Check**

Generate a cited recap and check its grounding.

Exclude later evidence. Say when there is not enough evidence.

**Review**

Read the timestamped recap and its trace.

Pause at a durable checkpoint; approve or reject, even after a restart.

A bounded Gemini planner chooses legal actions. Coverage gaps and provider failures remain visible.

4 / TECHNOLOGY AND LEARNING

# Each tool has a clear job.

Five course themes applied to one product. The detailed technology map is in docs/technology-map.md.

|                     |                                                 |                                                                                                                    |
|---------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Week 1 / App        | Streamlit, Python and persistent accounts.      | Save videos, queue jobs and review results. TypeScript/Zod powers the optional evidence demo.                      |
| Week 2 / Retrieval  | Gemini video, learned embeddings and citations. | Inspect a fixed interval, retrieve evidence, generate a recap and check the cited observations.                    |
| Week 3 / Workflow   | LangGraph, SQLite jobs and durable review.      | Bound model decisions, retry calls and resume review. Weighted LRU remains in the evidence demo.                   |
| Week 4 / Evaluation | Tests, Braintrust spans and a Nebius judge.     | Hosted traces work. Independent video labels and judge calibration remain open.                                    |
| Week 5 / LoRA       | PyTorch, Transformers, PEFT and scikit-learn.   | Train, evaluate and merge a BERT-tiny adapter. This adapts the technique, not the exact Qwen3/LLaMA Factory setup. |

Live capture uses FFmpeg and yt-dlp. Public deployment scaffolding is prepared; the current scope is local.

5 / EVIDENCE THAT IT WORKS

# Small tests. Clear limits.

These measurements use authored synthetic data. They do not establish real-video summary quality.

|                                                                                                                                                                                                              |                                              |                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|
| <div>38 passed</div> <div>Python product + evaluation tests</div>                                                                                                                                            | <div>40 / 40</div> <div>Evidence cases</div> | <div>Passed</div> <div>Streamlit + API flows</div> |
| <div>Spoiler test</div> <div>Request 00:00-15:00 of fictional Canyon Relay.</div> <div>Two ridge reports disagree. The update at 15:30 stays out.</div>                                                      |                                              |                                                    |
| <div>Evidence result</div> <div>Naive baseline 20/40 -&gt; workflow 40/40.</div> <div>Strict time and availability rules explain the measured improvement.</div>                                             |                                              |                                                    |
| <div>Separate LoRA lab</div> <div>52.5% baseline -&gt; 70% held-out accuracy.</div> <div>144 training / 40 held-out examples. Separate local classifier; the video agent uses a bounded model planner.</div> |                                              |                                                    |

Real Gemini check: small uploaded video, embeddings, cited recap and durable review passed.

YouTube and large-file processing remain provider-dependent. Active live capture and race accuracy need validation.

6 / RUN IT AND EXTEND IT

# A simple demo. A clear next step.

Video workspace: Python 3.11+ and a Gemini key. The optional evidence demo also needs Node.js 22.13+.

START LOCALLY

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
# configure Gemini in .env
python scripts/run_product.py
```

Open <http://localhost:8501>

Try this first

Create a local account. Save a short video and queue a question. Open Jobs / results, check timestamps and approve or reject the recap.

For a key-free fallback, use run\_demo.py and the fictional Canyon Relay scenario. Setup steps are in docs/setup.md.

|      |                                                |                                                                                                            |
|------|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Next | Validate real race clips and an active stream. | Review claims independently, calibrate the judge and measure quality on varied sources.                    |
| Then | Pilot with race fans and organizers.           | Measure time saved and evidence accuracy. Explore timing feeds, runner watchlists and multi-camera recaps. |

[github.com/sivalinb/racetime-copilot](https://github.com/sivalinb/racetime-copilot)

Read: README -> docs/learning-guide.md -> docs/demo-guide.md. Course scope: user-provided Week 1-5 handouts.